Energy & Sustainability Dashboard

BI Portfolio Project - Decarbonization & Renewables Snapshot

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What this project demonstrates
• Goal: communicate decarbonization progress using an interactive, Power BI-like dashboard.
• KPIs: CO_2 per capita trend, renewables share trend, and GDP vs CO_2 relationship (with renewables as bubble size).
• Outcome: a shareable HTML dashboard suitable for GitHub Pages hosting.

Executive summary

This portfolio project is designed to look and feel like a BI dashboard you might build in Power BI: it is interactive, visually consistent, and optimized for storytelling. The dashboard supports a country filter and lets the user compare emissions and renewables trajectories, while also exploring the 2024 relationship between income level and CO_2 intensity.

Key insights surfaced in the dashboard

• **Not all high-income economies are high-emitting:** 2024 GDP per capita and CO_2 per capita show meaningful dispersion, highlighting the role of policy, energy mix, and efficiency.
• **Renewables share tends to trend upward** across countries in the demo, enabling a narrative around policy acceleration and technology adoption.
• **CO_2 per capita trends differ by development pathway:** some countries show declines, while others increase alongside industrialization.

Metrics & definitions

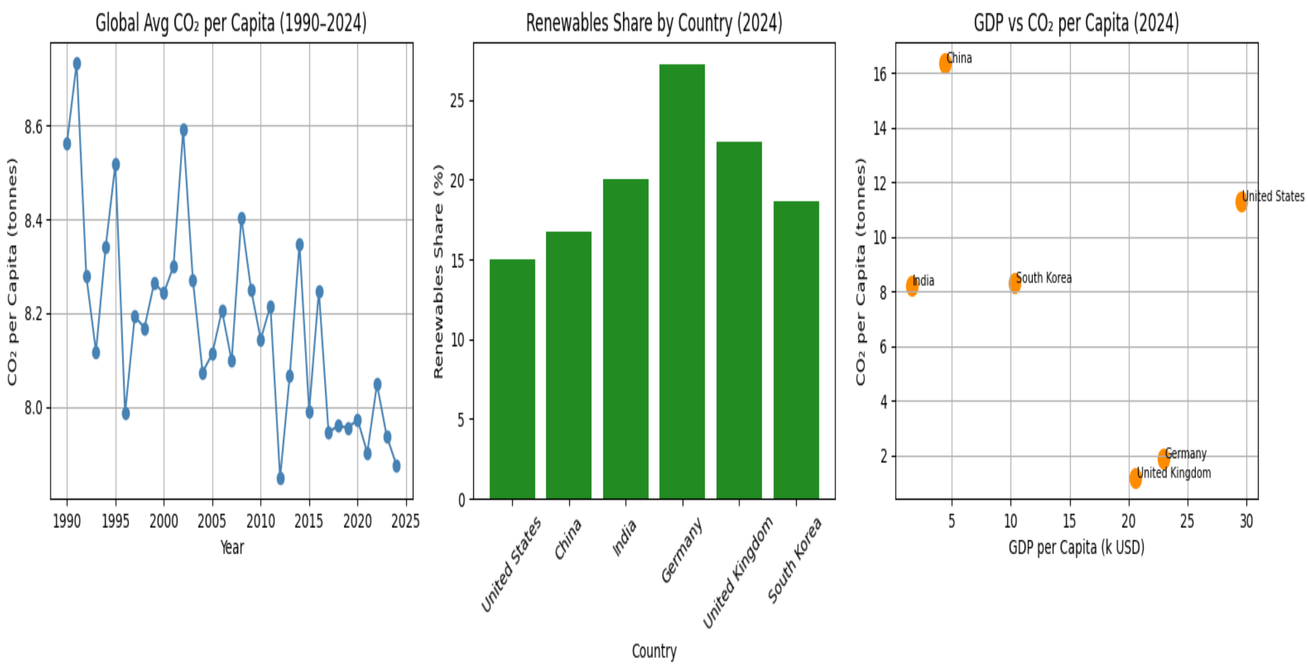
• **CO_2 per capita (t)** - annual CO_2 emissions divided by population (tons per person).
• **Renewables share (%)** - share of renewables in total energy (illustrative).
• **GDP per capita (k USD)** - economic output per person (thousands of USD).
• **Bubble size** - renewables share in 2024 (scaled for readability).

Dashboard walkthrough

The dashboard contains three coordinated views. Use the drop-down at the top to filter by country or compare all countries at once. Hover on any series/point to see exact values.

View	What it answers
1) CO2 per Capita by Country	Time-series to understand long-term emissions intensity. Useful for spotting structural br
2) Renewables Share by Country	Time-series to show momentum in renewable adoption (policy and technology progress)
3) GDP vs CO2 per Capita (2024)	Cross-sectional view for 2024 to compare economic development vs emissions intensity

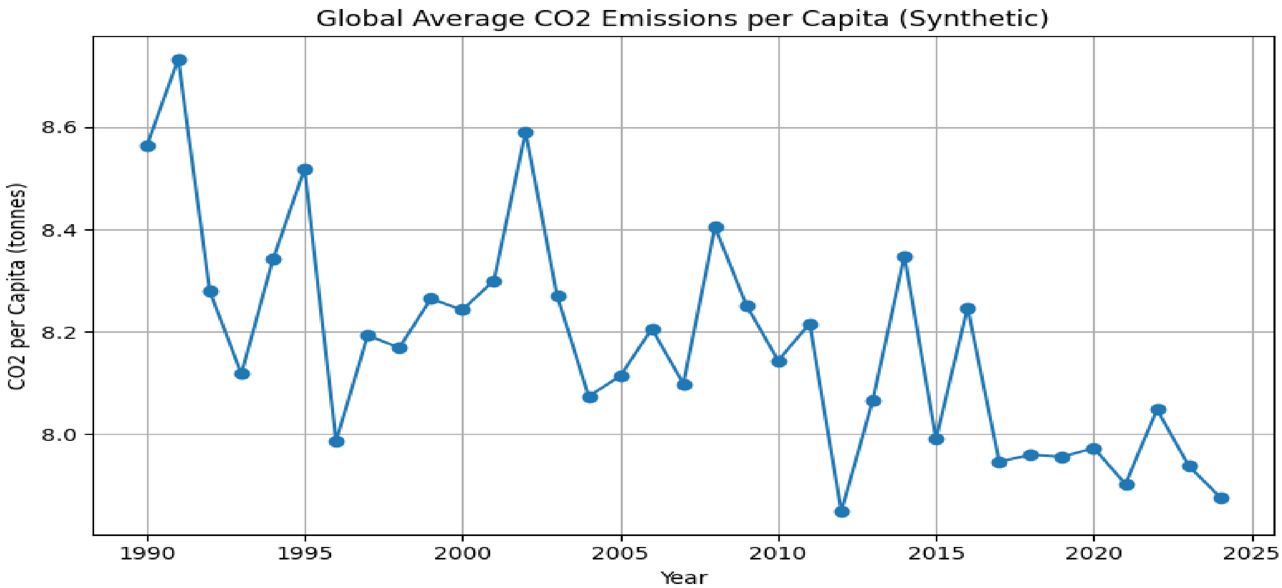
Dashboard preview (static snapshot)



Evidence of insight via visuals

Chart 1 - CO2 per capita trend (illustrative)

This view supports a narrative around decarbonization over time. In a real dataset, you would annotate major events (e.g., policy changes, energy shocks) and look for trend breaks.



Comparative views (2024 snapshot)

Chart 2 - Renewables share by country (2024)

This bar chart makes it easy to identify leaders/laggards. In a portfolio interview, explain how you would drill down by region or energy mix (wind/solar/hydro).

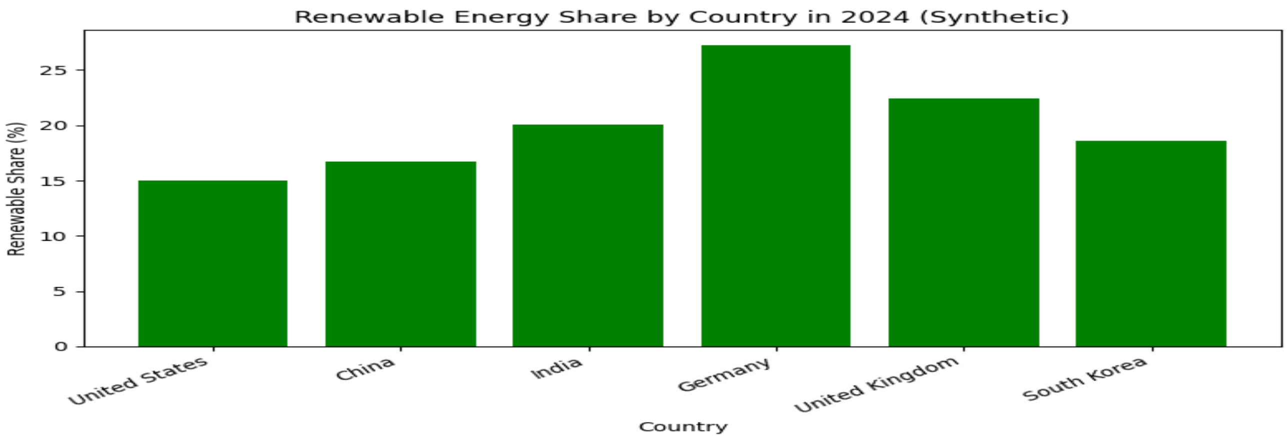
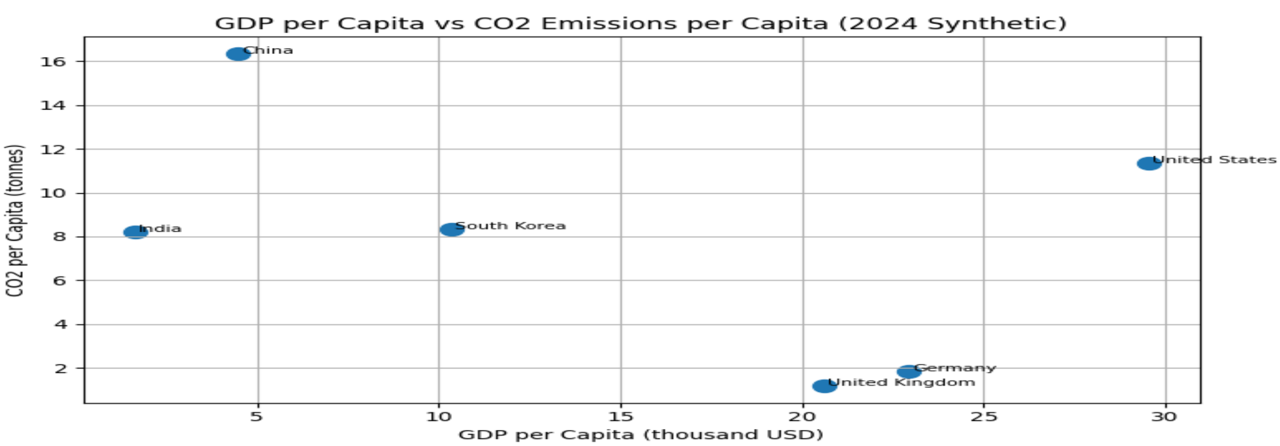


Chart 3 - GDP vs CO2 per capita (2024)

The scatter plot illustrates that income does not uniquely determine emissions intensity. Bubble size encodes renewables share to support a multi-metric story in one view.



Insights & recommendations

Insight 1 - Income does not uniquely determine emissions intensity
In the 2024 scatter plot, countries with similar GDP per capita can have very different CO_2 per capita. A BI takeaway is to segment by energy mix, industrial structure, and policy environment rather than relying on income as a proxy.

Insight 2 - Renewables adoption enables a narrative of transition speed
Renewables share trends upward across countries in the demo. Operationally, this KPI is useful for tracking targets (e.g., 2030 goals) and monitoring whether progress is accelerating or stalling.

Insight 3 - Trend context matters (levels vs trajectories)
CO_2 per capita levels can remain high even when a country is decarbonizing (declining trend). The dashboard therefore emphasizes both the level and the direction of change over time.

Recommended next steps (for a real-world portfolio upgrade)

• Swap the demo dataset with public sources (e.g., Our World in Data CO_2 dataset, IEA, World Bank indicators).
• Add a regional drill-down (EU, OECD, ASEAN) and a "baseline vs target" KPI card for 2030/2050 pathways.
• Implement cross-filtering: click a point/series to highlight the same country across all charts.
• Publish with GitHub Pages and include a short video walkthrough (30-60s) in the README.

Data note

This report uses a small demonstration dataset to showcase BI storytelling and dashboard craftsmanship. For production-grade analysis, replace the demo data with a public, auditable dataset and document the transformation steps.

References (recommended for the write-up)

• GitHub Pages docs - Configuring a publishing source: https://docs.github.com/en/pages/getting-started-with-github-pages/configuring-a-publishing-source-for-your-github-pages-site
• GitHub Pages Quickstart: https://docs.github.com/en/pages/quickstart